Zero-Shot Evaluation of gemini-3.5-flash on Intent Detection

A Case Study within the AfroBench-Lite Intent Detection Benchmark

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Abstract

This report presents a zero-shot evaluation of gemini-3.5-flash on intent detection within AfroBench-Lite, using the aggregated results in intent_detection_results.csv and the instance-level predictions in intent_detection_combined_results.csv. Across 14 languages and five prompt templates, the model reaches a macro-average F1 of 89.97% when selecting the best prompt per language, with prompt_5 as the strongest single fixed prompt overall (89.68% F1). The report summarizes the best-performing prompt per language and the per-prompt macro-average scores.

1 Background

This report concerns the intent-detection experiment in AfroBench-Lite. The task asks the model to classify a user utterance into one of the supported intent classes, such as alarm, balance, book_flight, or weather. The evaluation is conducted across 14 languages and five prompt templates, with the best prompt reported per language following the benchmark convention.

2 Data and Methodology

The evaluation uses two files in the experiment directory:

- intent_detection_results.csv: aggregated per-language and per-prompt F1/accuracy results.
- intent_detection_combined_results.csv: full sample-level predictions and outputs used to compute the aggregate metrics.

The model and task were configured as follows:

- Model evaluated: gemini-3.5-flash, accessed via the Gemini API.
- Task: Intent detection, using the AfroBench-Lite intent task configuration.
- Languages (14): eng, amh, hau, ibo, kin, lin, lug, orm, sna, sot, swa, xho, yor, zul.
- Shot setting: Zero-shot.
- Prompts: Five prompt templates (prompt_1–prompt_5), evaluated independently.
- Metrics: F1 and accuracy, reported both per language and per prompt.

3 Results

3.1 Best-Performing Prompt per Language

Table 1 reports the strongest prompt (by F1) for each of the 14 languages.

Table 1: Best-performing prompt per language for gemini-3.5-flash on intent detection (zero-shot).

Language	Best Prompt	F1	Accuracy
Amharic (amh)	prompt_3	0.9297	0.9297
English (eng)	prompt_5	0.8875	0.8875
Hausa (hau)	prompt_1	0.9625	0.9625
Igbo (ibo)	prompt_5	0.9313	0.9313
Kinyarwanda (kin)	prompt_4	0.8266	0.8266
Lingala (lin)	prompt_4	0.9266	0.9266
Luganda (lug)	prompt_4	0.9031	0.9031
Oromo (orm)	prompt_2	0.8719	0.8719
chiShona (sna)	prompt_2	0.9609	0.9609
Sesotho (sot)	prompt_5	0.7922	0.7922
Kiswahili (swa)	prompt_5	0.9359	0.9359
isiXhosa (xho)	prompt_1	0.9063	0.9063
Yoruba (yor)	prompt_2	0.8953	0.8953
isiZulu (zul)	prompt_3	0.8656	0.8656
Macro-average (14 languages)	—	0.8997	0.8997

3.2 Per-Prompt Macro-Average

Table 2 reports the macro-average F1 and accuracy across all 14 languages when each prompt is held fixed.

Table 2: Macro-average F1/accuracy across all 14 languages, per fixed prompt.

prompt_1	0.8942	0.8942
prompt_2	0.8941	0.8941
prompt_3	0.8938	0.8938
prompt_4	0.8954	0.8954
prompt_5	0.8968	0.8968

4 Discussion

The strongest single fixed prompt is prompt_5, which slightly outperforms the other templates on the averaged language-level results. The best prompt per language varies across templates, which suggests that prompt choice is still relevant for some languages even though the overall average is high. The aggregate summary in intent_detection_results.csv is supported by the sample-level predictions in intent_detection_combined_results.csv, which provide the full per-example evidence for the reported metrics.

5 Conclusion

The intent-detection experiment shows strong zero-shot performance for gemini-3.5-flash across the 14 AfroBench-Lite languages, with a macro-average F1 of 89.97% when selecting the best prompt per language. Prompt_5 is the best single fixed prompt overall, while the best prompt per language varies by language.